EE1101: Signals and Systems

Tutorial 7 Solutions

Solution 1

We use the analysis equation to find the Fourier transform of signals $x_1(t)$ and $x_2(t)$.

a) Here,

$$x_1(t) = \begin{cases} 4, & 0 < t < 1 \\ 2, & 1 < t < 2 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} X_1(j\omega) &= \int_{-\infty}^{\infty} x_1(t) e^{-j\omega t} dt \\ &= \int_0^1 4e^{-j\omega t} dt + \int_1^2 2e^{-j\omega t} dt \\ &= \frac{4(e^{-j\omega} - 1)}{-j\omega} + \frac{2(e^{-j2\omega} - e^{-j\omega})}{-j\omega} \\ &= 4e^{-j\omega/2}\left(\frac{e^{j\omega/2} - e^{-j\omega/2}}{j\omega}\right) + \\ &\quad 2e^{-j3\omega/2}\left(\frac{e^{j\omega/2} - e^{-j\omega/2}}{j\omega}\right) \\ &= 4\frac{\sin(\pi(\omega/2\pi))}{\pi(\omega/2\pi)} e^{-j\omega/2} + \\ &\quad 2\frac{\sin(\pi(\omega/2\pi))}{\pi(\omega/2\pi)} e^{-j3\omega/2} \\ &= 4\,\mathrm{sinc}(\omega/2\pi)e^{-j\omega/2} + 2\,\mathrm{sinc}(\omega/2\pi)e^{-j3\omega/2} \end{aligned}$$

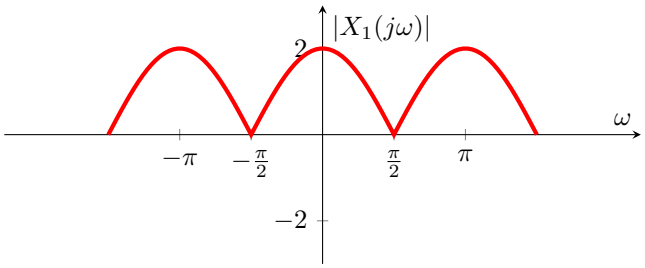
b) $x(t) = \begin{cases} \frac{|t|}{\tau} & |t| \le \tau \\ 0 & |t| > \tau \end{cases}$

$$\begin{aligned} X_2(j\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \\ &= \int_{-\tau}^{\tau} \frac{|t|}{\tau} e^{-j\omega t} dt \\ &= \int_{-\tau}^{\tau} \frac{|t|}{\tau} cos(\omega t) dt + j \times \int_{-\tau}^{\tau} \frac{|t|}{\tau} sin(\omega t) dt \\ &= \frac{2}{\tau}\int_0^{\tau} t\cos\omega t dt \\ &= \frac{2}{\tau}\left[\frac{t\sin\omega t}{\omega} + \frac{\cos\omega t}{\omega^2}\right]_0^{\tau} \\ &= 2\left[\frac{\tau\sin\omega\tau}{\omega\tau} + \frac{\cos\omega\tau - 1}{\omega^2}\right] \\ &= 2\tau\,\mathrm{sinc}(\omega\tau/\pi) - \tau\,\mathrm{sinc}^2(\omega\tau/2\pi) \end{aligned}$$

Solution 2

a) Let $x_1(t) = \delta(t+1) + \delta(t-1)$, then the Fourier transform is given by,

$$\begin{aligned} X_1(j\omega) &= \int_{-\infty}^{\infty} [\delta(t+1) + \delta(t-1)]\, e^{-j\omega t} dt \\ &= e^{j\omega} + e^{-j\omega} = 2\cos\omega \end{aligned}$$

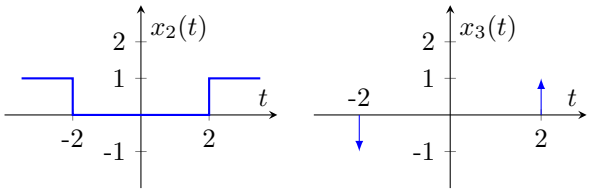


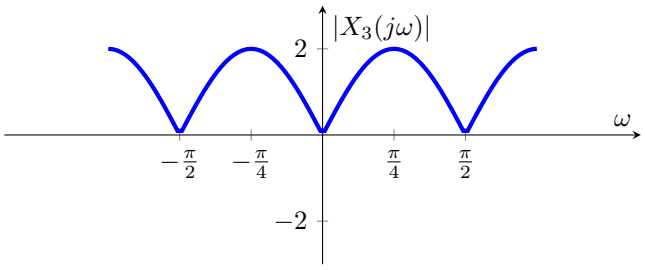
b)

$$\begin{aligned} x_3(t) = \frac{dx_2(t)}{dt} &= \frac{d}{dt}\left[u(t-2) + u(-2-t)\right] \\ &= \frac{d}{dt}\left[u(t-2) + u(-(t+2))\right] \\ &= \delta(t-2) - \delta(-(t+2)) \\ &= \delta(t-2) - \delta(t+2) \end{aligned}$$

Therefore,

$$\begin{aligned} X_3(j\omega) &= \int_{-\infty}^{\infty} [\delta(t-2) - \delta(t+2)]\, e^{-j\omega t} dt \\ &= e^{-j2\omega} - e^{j2\omega} = -2j\sin(2\omega) \end{aligned}$$





Solution 3

a)

$$x(t) = e^{-\frac{|t|}{2}} = \begin{cases} e^{\frac{t}{2}}, & t < 0 \\ e^{-\frac{t}{2}}, & t \geq 0 \end{cases}$$

Fourier transform,

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \\ &= \int_{-\infty}^{0} e^{\frac{t}{2}}e^{-j\omega t}dt + \int_{0}^{\infty} e^{-\frac{t}{2}}e^{-j\omega t}dt \\ &= \frac{1}{(0.5 - j\omega)}\left[e^{(0.5-j\omega)t}\right]_{-\infty}^{0} \\ &- \frac{1}{(0.5 + j\omega)}\left[e^{-(0.5+j\omega)t}\right]_{0}^{\infty} \\ &= \frac{1}{(0.5 - j\omega)} + \frac{1}{(0.5 + j\omega)} \\ X(j\omega) &= \frac{1}{0.25 + \omega^2} \end{aligned}$$

b) Given

$$x(t) = \sin(2\pi t)e^{-t}u(t) = \frac{(e^{j2\pi t} - e^{-j2\pi t})}{2j}e^{-t}u(t)$$

$$\begin{aligned} X(j\omega) &= \int_0^{\infty} \frac{(e^{j2\pi t} - e^{-j2\pi t})}{2j}e^{-t}e^{-j\omega t}dt \\ &= \frac{1}{2j}\left\{\int_0^{\infty}(e^{t(j2\pi - j\omega - 1)}dt - \int_0^{\infty}(e^{-t(j2\pi + j\omega + 1)}dt\right\} \\ &= \frac{1}{2j}\left\{\frac{(-e^{t(j2\pi - j\omega - 1))}}{(1 - j(2\pi - \omega))}\Big|_0^{\infty} - \frac{(-e^{-t(j2\pi + j\omega + 1))}}{(1 + j(2\pi + \omega))}\Big|_0^{\infty}\right\} \\ &= \frac{1}{2j}\left\{\frac{1 + j(2\pi + \omega) - 1 + j(2\pi - \omega)}{(1 + j\omega)^2 + 4\pi^2}\right\} \\ &= \frac{2\pi}{(1 + j\omega)^2 + 4\pi^2} \end{aligned}$$

Solution 4

a) The $x(t)$ plot is as shown in figure 1.
Using the Fourier transform equation, we have

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

$$X(j\omega) = \int_{-1}^{1} \frac{(t+1)}{2}e^{-j\omega t}dt = \int_{-1}^{1}\left[\frac{te^{-jwt}}{2} + \frac{e^{-j\omega t}}{2}\right]dt$$

And integrate the first term by parts

$$= \frac{1}{2}\left(\left[\frac{te^{-j\omega t}}{-j\omega} + \frac{e^{-j\omega t}}{\omega^2}\right]_{-1}^{1} + \left[\frac{e^{-jwt}}{-j\omega}\right]_{-1}^{1}\right)$$

$$= \frac{e^{-j\omega}}{-j\omega} - \frac{\sin(\omega)2j}{2\omega^2}$$

$$X(j\omega) = \frac{e^{-j\omega}}{-j\omega} + \frac{\sin(\omega)}{j\omega^2}$$

b) Real part of $X(j\omega) = Real\left(\frac{e^{-j\omega}}{-j\omega} + \frac{\sin(\omega)}{j\omega^2}\right) = Real\left(\frac{1}{\omega}\sin(\omega) + \frac{j}{\omega}\cos(\omega) - \frac{j}{\omega^2}\sin(\omega)\right) = \frac{\sin\omega}{\omega}$

& Even part of $x(t)$ is given as $\Rightarrow \frac{x(t) + x(-t)}{2} = \frac{(t+1)/2 + (-t+1)/2}{2} = 0.5$, in the domain $t = [-1, 1]$.

The Even($x(t)$) plot is as shown in figure 2.

$X(j\omega) = \int_{-1}^{1} 0.5.e^{-j\omega t}dt = \frac{1}{2}\frac{e^{-j\omega t}|_{-1}^{1}}{-j\omega} = \frac{\sin\omega}{\omega}$

And thus they are equal.

c) Odd part of $x(t)$ is given as $\Rightarrow \frac{x(t) - x(-t)}{2} = \frac{(t+1)/2 - (-t+1)/2}{2} = t/2$, in the domain $t = [-1, 1]$.
The Odd($x(t)$) plot is as shown in figure 3.

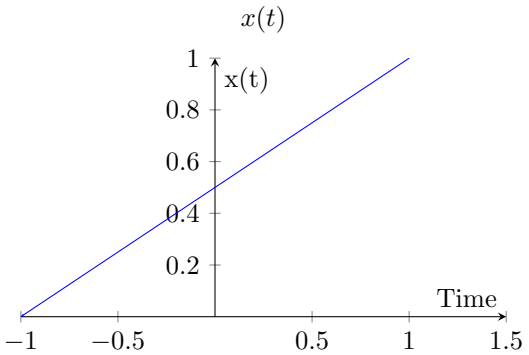
$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$

$X(j\omega) = \int_{-1}^{1} \frac{t}{2}e^{-j\omega t}dt = \frac{-j\sin(\omega)}{\omega^2} + \frac{j\cos(\omega)}{\omega}$

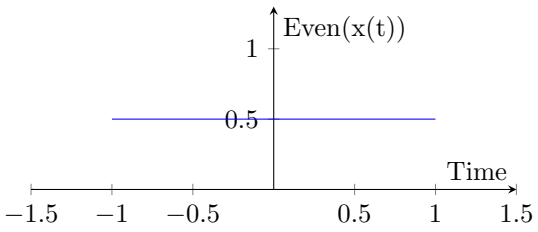
Property: The fourier transform of the odd part of $x(t)$ is the same as j times imaginary part of the answer to part (a). i.e

Let Y= $Im\left[\frac{e^{-j\omega}}{-j\omega} + \frac{\sin(\omega)}{j\omega^2}\right] = \frac{-\sin\omega}{\omega^2} + \frac{\cos\omega}{\omega}$

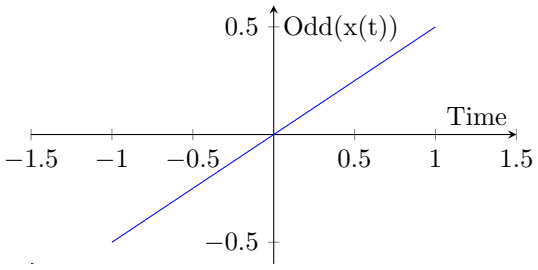
now $jY = \frac{-j\sin\omega}{\omega^2} + \frac{j\cos\omega}{\omega}$



$$\frac{x(t)+x(-t)}{2}$$



$$\frac{x(t)-x(-t)}{2}$$



Solution 5

a) $x(t) = \begin{cases} 1-|t|, & |t| \le 1 \\ 0, & |t| > 1 \end{cases}$

F.T,

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \\ &= \int_{-1}^{0}(1+t)e^{-j\omega t}dt + \int_{0}^{1}(1-t)e^{-j\omega t}dt \\ &= \int_{0}^{1}(1-t)e^{j\omega t}dt + \int_{0}^{1}(1-t)e^{-j\omega t}dt \\ &= 2\int_{0}^{1}(1-t)\cos\omega t dt \\ &= 2\left[\frac{(1-t)\sin\omega t}{\omega} - \frac{\cos\omega t}{\omega^2}\right]_0^1 \\ &= \frac{2}{\omega^2}(1-\cos\omega) = \frac{4\sin^2\left(\frac{\omega}{2}\right)}{\omega^2} \\ &= \operatorname{sinc}^2\frac{\omega}{2\pi} \end{aligned}$$

b) Let $Let, y(t) = x(t+T) + x(t-T) = x_1(t) + x_2(t)$

F.T,

$$X_1(j\omega) = \int_{-\infty}^{\infty} x_1(t)e^{-j\omega t}dt = \int_{-\infty}^{\infty} x(t+T)e^{-j\omega t}dt$$

Take $t+T = z \implies dt = dz$

$$X_1(j\omega) = \int_{-\infty}^{\infty} x(z)e^{-j\omega(z-T)}dz = X(j\omega)e^{j\omega T}$$

Similarly, $X_2(j\omega) = X(j\omega)e^{-j\omega T}$,

Therefore,

$$\begin{aligned} Y(j\omega) &= X_1(j\omega) + X_2(j\omega) \\ &= 2X(j\omega)\left[\frac{e^{j\omega T}+e^{-j\omega T}}{2}\right] \\ &= 2X(j\omega)\cos(\omega T) \end{aligned}$$

c)

$$y(t) = x(t+3) + x(t-3)$$

Therefore, using result from Qn.5(b),

$$Y(j\omega) = 2X(j\omega)\cos(3\omega)$$

Solution 6

a) If $x(t)$ is even in t, $x(t) = x(-t)$,

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \\ &= \int_{-\infty}^{0} x(t)e^{-j\omega t}dt + \int_{0}^{\infty} x(t)e^{-j\omega t}dt \end{aligned}$$

In first part replace $t \to -a \implies dt = -da$

$$\begin{aligned} &= \int_{\infty}^{0} x(-a)e^{j\omega a}(-da) + \int_{0}^{\infty} x(t)e^{-j\omega t}dt \\ &= \int_{0}^{\infty} x(-a)e^{j\omega a}da + \int_{0}^{\infty} x(t)e^{-j\omega t}dt \end{aligned}$$

In first part replace $a \to t \implies da = dt$

$$\begin{aligned} &= \int_{0}^{\infty} x(-t)e^{j\omega t}dt + \int_{0}^{\infty} x(t)e^{-j\omega t}dt \\ &= \int_{0}^{\infty} (x(-t)e^{j\omega t} + x(t)e^{-j\omega t})dt \\ &= \int_{0}^{\infty} (x(t)\left[e^{j\omega t} + e^{-j\omega t}\right])dt \\ &\because x(t) = x(-t) \\ &= 2\int_{0}^{\infty} x(t)\cos(\omega t)dt \end{aligned}$$

b) If $x(t)$ is odd in t, $x(t) = -x(-t)$,

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \\ &= \int_{-\infty}^{0} x(t)e^{-j\omega t}dt + \int_{0}^{\infty} x(t)e^{-j\omega t}dt \end{aligned}$$

In first part replace $t \to -a \implies dt = -da$

$$\begin{aligned} &= \int_{\infty}^{0} x(-a)e^{j\omega a}(-da) + \int_{0}^{\infty} x(t)e^{-j\omega t}dt \\ &= \int_{0}^{\infty} x(-a)e^{j\omega a}da + \int_{0}^{\infty} x(t)e^{-j\omega t}dt \end{aligned}$$

In first part replace $a \to t \implies da = dt$

$$\begin{aligned} &= \int_{0}^{\infty} (x(-t)e^{j\omega t} + x(t)e^{-j\omega t})dt \\ &= \int_{0}^{\infty} (x(t)\left[-e^{j\omega t} + e^{-j\omega t}\right])dt \\ &\because x(t) = -x(-t) \\ &= -2j\int_{0}^{\infty} x(t)\sin(\omega t)dt \end{aligned}$$

Solution 7

a) The inverse fourier transform is

$x_1(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}[2\pi\delta(\omega) + \pi\delta(\omega - 4\pi) + \pi\delta(\omega + 4\pi)]e^{j\omega t}d\omega$

$= \frac{1}{2\pi}[2\pi e^{j0t} + \pi e^{j4\pi t} + \pi e^{-j4\pi t}]$

$= 1 + \frac{1}{2}e^{j4\pi t} + \frac{1}{2}e^{-j4\pi t} = 1 + cos(4\pi t)$

b) $x_2(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_2(j\omega)e^{j\omega t}d\omega$

$= \frac{1}{2\pi}\int_0^2 2e^{j\omega t}d\omega + \frac{1}{2\pi}\int_{-2}^{0}(-2)e^{j\omega t}d\omega$

$= \frac{1}{\pi jt}(e^{j2t} - 1) - \frac{1}{\pi jt}(1 - e^{-j2t})$

$= \frac{2(\cos(2t)-1)}{\pi jt}$

$= \frac{4j\sin^2 t}{\pi t}$